

Joseph Sandoval

jrs004@ucsd.edu | linkedin.com/in/joseph-r-sandoval | github.com/Jtoast65 | (310) 483-0596 | San Diego, CA | U.S. Citizen

Education

University of California, San Diego

B.S. Data Science, Minor in Business - Provost Honors

La Jolla, CA

Expected June 2027

Relevant Coursework: Data Structures, Machine Learning, Data Visualization, Probability & Statistics, Linear Algebra

University of California, Merced

Coursework toward B.S. Computer Science & Engineering, transferred to UCSD

Merced, CA

August 2022 - May 2024

Skills

Programming Languages: Python, SQL, C++, C, JavaScript, R

ML/Data Science: XGBoost, PyTorch, TensorFlow, scikit-learn, Pandas, NumPy, Matplotlib, OpenCV, NLTK, PIL

GenAI/LLM: retrieval-augmented generation (RAG), OpenAI embeddings (text-embedding-3-small), gpt-4o-mini, pgvector (HNSW), semantic search, prompt grounding

Backend/Frontend: FastAPI, Pydantic, React, React Router, Streamlit

Data/Infrastructure: PostgreSQL, AWS (RDS, S3), Apache Airflow, dbt, PySpark, Supabase, MongoDB, MySQL, Docker, Render, Git, Jupyter, Power BI

Projects

NFL Win Probability, RAG Q&A & Orchestration Pipeline

Python, XGBoost, FastAPI, Airflow, dbt, AWS

Repo: github.com/Jtoast65/NFL-Analytics | *Dashboard:* nfl-analytics-crwjgsvzwmwtrhl75mwpmtpt.streamlit.app | *API:* nfl-analytics-api-aku6.onrender.com/docs

- Built an end-to-end analytics pipeline ingesting 407K+ play-by-play records across 10 NFL seasons (2016-2025) into a normalized five-table PostgreSQL schema with idempotent ingestion.
- Trained and calibrated a 14-feature XGBoost win probability model with isotonic regression under a strict temporal split (train 2016-2022, validate 2023, test 2024), reaching a 0.140 Brier score and 0.884 ROC-AUC and beating the public nflfastR baseline by 8%.
- Added a retrieval-augmented Q&A layer over ~7,600 embedded documents using OpenAI text-embedding-3-small and gpt-4o-mini grounded strictly in retrieved context, served from a pgvector HNSW store through a FastAPI endpoint and a Streamlit page.
- Orchestrated the batch pipeline as a five-task Apache Airflow DAG (ingest, S3 raw lake, dbt build, model retrain, RAG reindex) running against AWS RDS and S3, with 15 dbt data-quality tests across three marts and a PySpark port of feature engineering verified row-for-row against pandas.

SoCalGuesser Image Classification

PyTorch

Repo: github.com/Jtoast65/SocalGeoGuesser

- Trained a MobileNetV3-Small transfer-learning classifier over six Southern California cities to 73.33% accuracy against a 16.7% random baseline, with stratified splits, AdamW, and multi-view test-time augmentation.

Unbeatable Tic-Tac-Toe AI

C++

Repo: github.com/Jtoast65/Unbeatable-Tic-Tac-Toe-AI

- Built a C++20 game-tree AI that expands future board states breadth-first through a hand-written doubly linked list and queue, then selects moves by minimax, with player-selectable board size and a 100K-1M node search budget trading search speed against decision quality.

Work Experience

Whole Foods Market

San Diego, CA

Produce Operations Associate, Part-time

June 2026 – Present

- Receive and break down produce pallet deliveries, organizing backroom storage to fit full shipments into limited cooler and floor space.
- Rotate stock on a FIFO basis to reduce spoilage and track departmental produce waste.

Associated Students of UC Merced

Merced, CA

Senator, School of Engineering

August 2023 – July 2024

- Managed and tracked a \$200K operating budget for student organizations and developed a \$240K fiscal plan to optimize funding allocation for the following year.
- Held office hours, allocated funds, and resolved conflicts between student organizations.

Certifications & Leadership

Certifications: Claude Code 101 - Anthropic (June 2026); Krach Leadership Certificate - Sigma Chi (July 2024) |

Leadership: Vice Chair - Budget & Finance and Sergeant at Arms (ASUCM); Intern Educator - Business Society; Data Science Student Society; ACM